

EXPERIMENT-1

8-BIT ADDITION

AIM: To write an assembly language program to implement 8-bit addition using 8085 processor.

ALGORITHM:

1. Start the program by loading the first data in the accumulator.
2. Move the data to a register
3. Get the second data and load it into the accumulator.
4. Add the two register contents
5. Check the carry.
6. Store the value of sum & carry in the memory location.
7. Halt.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 8050	load accumulator with first number in the address 8050
MOV B, A	Move the data from accumulator to 'B' register.
LDA 8051	load accumulator with second number in the address 8051
ADD B	Add the data of 'B' register with accumulator
STA 8052	Store the data (output) of the accumulator in address 8052
HLT	Halt.

INPUT:

ADDRESS	DATA
8050	1
8051	2

OUTPUT:

ADDRESS	DATA.
	3

Result:

Thus the program was executed successfully using 8085 processor simulator.

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A07

BC0500

DE0000

HL0000

PSW0000

PC420C

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

8050-+05

Update Memory

Load me at

1LDA 8050

2MOV B,A

3LDA 8051

4ADD B

5STA 8052

6HLT

7

8

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

1F7280505

1F7380512

1F7480527

1F7580530

1F7680540

1F7780550

1F7880560

1F7980570

1F7A80580

1F7B80590

1F7C80600

1F7D80610

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle